Vishwas Tiwari AI/ML & Backend Engineer

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SUMMARY

Focused on **AI/ML and Backend systems**, but I don't stay in a box. I build with **ML models**, **design backend flows**, **automate testing** and **deployment**, **experiment with NLP**, and make different technologies work together. Adaptability matters more than specialization today. Always learning, building, and turning ideas into something real.

EXPERIENCE

AI/ML & Backend Engineer - **TestDino**

May 2026 - Present

AI/ML and Backend Engineer specializing in AI product development, LLM integrations, and backend infrastructure. Experienced in building production-ready AI features, integrating enterprise tools, developing MCP-based AI systems, and optimizing distributed backend flows. Passionate about leveraging AI and scalable engineering practices to improve automation, developer productivity, and software quality.

- Developed multiple integrations for TestDino including Slack, GitHub, JIRA, and Claude Web to improve automation and developer assitant and workflows.
- Built the MCP (Model Context Protocol) infrastructure for TestDino, enabling multiple AI clients to securely interact with TestDino report data and enhance QA automation workflows.
- Worked on backend architecture improvements including server load balancing, database sharding, and CI/CD pipeline optimization to improve scalability and system reliability.

Data Science Intern - **Alphabin Tech**

Aug 2025 - April 2026

Data Science Intern experienced in Machine Learning, NLP, LLM evaluation, text classification, clustering, and AI workflow automation. Built practical solutions for model testing, error analysis, and process automation using modern AI technologies.

- Built LLM testing workflows using Playwright and DeepEval to evaluate AI model accuracy, response structure, and resilience against prompt injection attacks.
- Developed a text classification and clustering model for automated error grouping from raw Playwright error logs, improving debugging and failure analysis workflows.
- Created an AI-driven blog publishing pipeline that automated content publishing workflows, reduced manual effort, and improved operational efficiency.

EDUCATION

Bhagwan Mahavir University, Surat **2023-26**

- Bachelor of Computer Applications (Web Development)
- CGPA: 7.5/10
- Developed "FuelMate", a full-stack fuel management platform as final-year project.
- Participated in university-level technical competitions

SKILLS

AI & Machine learning

- Python, Scikit-learn, TensorFlow
- Machine Learning (Supervised), Deep Learning
- NLP, Clustering, Classification, Neural Networks
- LLM Integrations, Prompt Engineering
- MCP, Agentic AI, Embeddings

Data & Analytics

- Pandas, NumPy, MatplotLib
- EDA, Data Visualization, Feature Engineering
- Data Handling and Pre-Processing

Backend Development

- JavaScript, Node.js, Express.js
- REST APIs, Authentication, Middlewares, Services, JWT, etc.
- Database scaling, Architecture Designing, Horizontal Scaling, Microservices.

DevOps & Infrastructure

- Docker
- GitHub Actions, Azure pipelines
- CI/CD, Load Balancing, CDN

Testing and Browser Automation

- Playwright
- DeepEval
- Test Automation, Shard Optimization, CI orchestration.

KEY PROJECTS

Neural network classification on MNIST with Scratch

Built a neural network from scratch in Python to classify handwritten digits from the MNIST dataset. Implemented core deep learning components including forward propagation, backpropagation, gradient descent, and ReLU activation without using high-level ML frameworks, achieving high test accuracy.

- 3-layer neural network from scratch in Python for MNIST digit classification.
- Implemented forward propagation, backpropagation, gradient descent, and ReLU activation using Python only.
- Trained and optimized the model over 900 iterations, achieving 85% test accuracy.
- Developed a deeper understanding of neural network fundamentals by avoiding high-level ML frameworks.

Fuelmate - On Demand Fuel Delivery System

Developed a full-stack MERN application for fuel ordering and delivery management, featuring role-based access for customers, fuel stations, delivery personnel, and administrators. Implemented secure authentication, real-time location tracking, and scalable backend services using modern web technologies and production-grade engineering practices.

- A full-stack MERN application using React, TypeScript, Node.js, Express, and MongoDB.
- Designed a role-based system supporting customers, fuel stations, delivery personnel, and administrators with dedicated dashboards and permissions.
- Implemented JWT authentication with httpOnly refresh tokens and Google OAuth for secure user access.
- Developed REST APIs for authentication, order management, and user management with complete CRUD functionality.
- Configured GitHub Actions CI/CD pipelines for automated build, testing, and deployment workflows.
- Deployed and maintained both frontend and backend services on Render.
- Implemented production-grade monitoring and logging using Winston with daily rotating log files.

Gold Price Predictor ML Model

Developed a machine learning model to predict gold prices using historical market indicators such as stock indices, oil prices, silver prices, and currency exchange rates. Performed exploratory data analysis, correlation analysis, feature selection, and model training using ensemble learning techniques to generate accurate price predictions.

- Developed a gold price prediction model using historical financial and commodity market data.
- Performed data preprocessing, correlation analysis, and feature selection to identify key price-driving factors.
- Trained and evaluated a Random Forest Regression model using Python and Scikit-learn.
- Analyzed relationships between gold prices and market indicators including stock indices, oil prices, silver prices, and EUR/USD exchange rates.
- Utilized Pandas, NumPy, Matplotlib, and Seaborn for data analysis and visualization.

Cafe Sales Insights

Conducted an end-to-end data analysis of café sales and customer behavior data to uncover revenue drivers, customer purchasing patterns, and operational optimization opportunities. Leveraged statistical analysis, visualization, and experimentation techniques to generate actionable business recommendations for pricing, marketing, staffing, and menu design.

- Analyzed café sales and customer behavior data to identify revenue trends, menu performance, and operational improvements.
- Applied statistical techniques including correlation analysis, regression modeling, and A/B testing to generate actionable business insights.
- Built dashboards and visualizations to communicate findings and support data-driven decision making.
- Recommended optimizations for pricing, staffing, marketing, and menu strategy based on analytical results.

This resume shows the highlights. The full story lives at: https://aboutvishwas.netlify.app